CURVE FITTING LEAST SQUARE METHOD

- **Curve Fitting: -**

(1) **Linear Curve Fitting**: Equation of Linear curve is $y = a + bx$, where a and b can be found from the following:

$\sum y = an + b\sum x$ and $\sum xy = a\sum x + b\sum x^2$

So, $b = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2}$, $\quad a = \dfrac{\sum y}{n} - b\dfrac{\sum x}{n}$.

(2) **Quadratic Curve Fitting**: Equation of Linear curve is $y = a + bx + cx^2$, where a, b and c can be found from the following:

$$\sum y = an + b\sum x + c\sum x^2$$
$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$$
$$\sum x^2 y = a\sum x^2 + b\sum x^3 + c\sum x^4 .$$

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